Project Proposal • September 28, 2026

Submitted by: Dimitri Chrysafis **Course:** COMPSCI 566

Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

Team Members

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Roles: baselines and evaluation (Dimitri Chrysafis); DFD, refinement, and webpage (Vincent Zhou). Both of us share the writing, experiments, and the presentation.

The Problem

Take a photo of a chair. Nothing in the pixels tells you whether it is a real chair two meters away or a toy chair on a desk, because shrinking a scene and moving it closer leaves the image almost unchanged. That is the scale ambiguity at the center of monocular depth estimation. Our goal is a dense depth map from one RGB image, and for a short focal stack, depth in real units.

Learned models cope by giving up on scale. They predict relative depth, defined only up to an affine transform of inverse depth, $z = 1/(a\,d + b)$, and their training treats blur as noise rather than signal. But defocus is the one monocular cue that carries absolute scale, because it comes straight from lens physics. Under the thin-lens model, a point at depth z produces a Gaussian point spread with standard deviation

$$\sigma(z) = k\left|1 - \frac{z_f}{z}\right|, \qquad k = \frac{f_{\text{px}}\, f}{2N\,(z_f - f)},$$

where f_{px} is the focal length in pixels, f the physical focal length, N the f-number, and z_f the focus distance. Measured blur is a direct function of depth in meters, and no other single-image cue gives that.

We picked this topic for two reasons. Depth turns ordinary cameras into 3D sensors (portrait mode, AR, drones, robot navigation), and the problem sits where the course's optics (image formation, focus and defocus) meets modern learned models. It also has a clean scoreboard: the standard metrics (AbsRel, RMSE, δ thresholds) mean improvements are measurable, not anecdotal.

State of the Art

Monocular depth is largely solved in the relative sense. MiDaS [1] and DPT [2] trained across mixed datasets with a scale-invariant loss; Depth Anything [3] scaled the recipe to 62M unlabeled images; Depth Anything V2 [4] trains on synthetic labels and ships models from 25M to 1.3B parameters, more than 10× faster than the diffusion-based Marigold [5]. ZoeDepth [6] adds metric heads, and Monodepth2 [7] is the standard self-supervised baseline. All of them output affine-invariant inverse depth, and none of them uses defocus as a cue.

Depth from defocus goes the other way. Pentland [8] and Subbarao [9] recover depth from local blur through the thin-lens model; deep DFD methods [10,11] learn the blur-to-depth mapping from focal stacks; SGM [12] instead uses geometry from two views. The catch is reliability. DFD

needs texture, a controlled stack, and good calibration, and it breaks down exactly where the learned models are strongest.

Family	Output	Weak spot
Monocular learned [1–4,6,7]	relative depth	no metric scale by construction
Depth from defocus [8–11]	metric depth	needs texture, a stack, calibration
Stereo matching [12]	metric depth	needs two views; weak on textureless regions

Our plan takes the dense prediction from the first row and the scale from the second, at test time, with no retraining.

Approach: Re-implement, Then Improve

Phase 1 is re-implementation, so every claim later has a number attached:

- DA-V2 (ViT-S, 25M) [4] zero-shot on NYU-Depth v2 and DDFF-12, scored after least-squares scale-and-shift alignment.
- Classical DFD from scratch: a focus measure (modified Laplacian and Tenengrad) over a focal stack, converted to depth with the thin-lens model [8,9], scored on DDFF-12.
- If time allows, an SGM stereo baseline [12] on the standard pairs.

Phase 2 is the new part. We estimate per-pixel blur $\hat{\sigma}$ from the focal stack (Laplacian energy across focus settings) and solve for a metric depth field

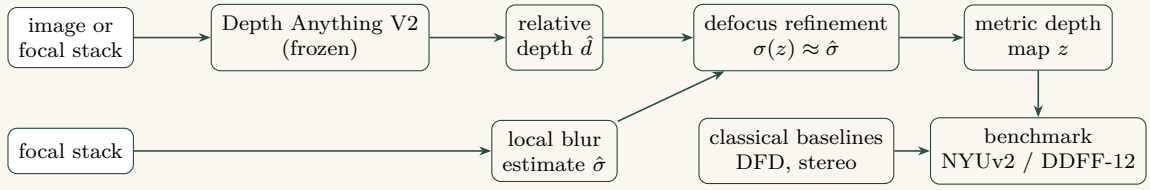
$$z = \frac{1}{s\,\hat{d} + t + r(\mathbf{x})},$$

an affine rescale of the network's relative depth $\hat{d}$ plus a smooth residual r, by minimizing

$$\sum_{\mathbf{x}} \rho\big(\sigma(z(\mathbf{x})) - \hat{\sigma}(\mathbf{x})\big) + \lambda \|\nabla r\|^2,$$

with ρ a Huber loss. The residual matters because one global scale cannot fix every region at once; r lets the blur data correct the network locally. The optimization runs per image at test time, a few hundred gradient steps, initialized from the network output and the DFD estimate. No retraining, no extra labels.

The two sources fail in opposite places. The network is dense and stable but scale-free; defocus is scale-aware but noisy and helpless on textureless surfaces. The network's output family (affine-invariant inverse depth) is exactly what blur constrains, so the fit is well-posed wherever blur is measurable. And if the refinement does not pan out, the benchmark and error analysis still stand on their own.



Relative depth from a frozen zero-shot model plus blur from the focal stack; the refinement enforces thin-lens consistency and outputs metric depth.

Evaluation and Plan

Metrics are AbsRel, SqRel, RMSE, RMSE-log, and the $\delta < 1.25, 1.25^2, 1.25^3$ thresholds; the refinement also reports metric scale error mean $|\log(z_{\text{pred}}/z_{\text{gt}})|$ and a blur-consistency error.
For data we use the 654-image NYU-Depth v2 test split [13], DDFF 12-Scene [10] for real focal stacks with ground truth, and our own phone captures for qualitative results.
We compare DA-V2 raw and affine-aligned, classical DFD, and the full refinement. Ablations drop the residual, drop the smoothness term, and vary the frame count (1, 3, 5) and λ. Since the refinement optimizes per image, we also report runtime. The final results will be a benchmark table of learned vs. classical depth, the refinement's gain in accuracy and scale error, error maps and side-by-side stacks, and a project webpage with code.
The concrete steps from here to December:

Dates	**Steps**
Sep 29	Proposal submitted.
Sep 30–Oct 6	DA-V2 running on NYUv2; scale-and-shift alignment script; first baseline numbers.
Oct 7–20	DFD baseline on DDFF-12; error analysis for both baselines.
Oct 21–29	Write the midterm report (due Oct 29); freeze Phase 1 numbers.
Oct 30–Nov 13	Implement the refinement; ablations on residual, smoothness, frames, λ.
Nov 14–24	Capture our own focal stacks; final experiments; build the webpage.
Nov 25–Dec 1	Prepare the presentation.
Dec 1–10	Present; finalize webpage and code (Dec 10).

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